User

:System (CineTrack)

:LocationManager (GPS Sensor)

:MapView (OSMDroid)

:FoursquareApiService (Retrofit)

:External Map App (Google Maps)

Scenario 1: Request Location & Render Interactive Map

1 openNearbyCinemas()

2 checkLocationPermission(ACCESS_FINE_LOCATION)

alt [Not Granted]

3 requestPermissionDialog()

4 grantPermission()

5 requestLocationUpdates() / getLastKnownLocation()

6 currentLocation(latitude, longitude)

7 readSearchRadiusFromPreferences()

8 setCenter(currentLocation) & setZoom(dynamicZoom)

9 drawRadiusCircumference(Polygon.pointsAsCircle, radiusMeters)

10 addMarker(userLocationMarker)

11 showLoadingIndicator(true)

12 searchNearbyCinemas(latLng, query, categories, radiusMeters)

13 GET places/search?ll=lat,lon&radius=...&sort=DISTANCE

14 placesJsonPayload

15 mapToCinemaEntities()

16 showLoadingIndicator(false)

alt [Cinemas Found in Radius]

17 showEmptyStateMessage("No cinemas found nearby")

[Found]

18 plotCinemaMarkers(cinemasList)

19 renderCinemasRecyclerView(cinemasList)

20 displayCinemasSplitView(map + list)

Scenario 2: Synchronized Map Focus & Navigation

alt [Taps Cinema in List]

21 tapCinemaCard(cinema)

22 animateTo(cinema.coordinates)

23 showInfoWindow(cinema.name, distance, address)

[Clicks Marker on Map]

24 clickMarker(cinema)

25 showPopupBubble(cinema.name, distance, address)

26 requestDirections(cinema)

27 startActivity(ACTION_VIEW, "geo:lat,lng?q=CinemaName")

28 showTurnByTurnNavigation()

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